

Thomson Tong

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EDUCATION

University of Waterloo - Biomedical Engineering

Candidate: BASc | Second-Year Student

September 2024 - Present

- **Relevant Coursework:** Data Structures & Algorithms, Materials Science Lab (software-oriented data preparation, analysis, visualization, and technical reporting using Python)

TECHNICAL EXPERIENCE

Full Stack Developer (Co-op) | Mission Perform

Jan 2026 – Present

- Shipped 15+ production features using the MERN stack on AWS for pharmaceutical manufacturing clients.
- Implemented a relational tagging architecture and re-architected the attendance subsystem, improving reliability by 50%.
- Reduced application errors by 40% by resolving high-volume support tickets and addressing production edge cases.
- Enforced data isolation via JWT-based role-based authentication and hierarchical access controls.

AI Researcher | WAT.ai - Waterloo, Ontario

June 2025 - December 2025

- Published an architecture-led analysis of multimodal vision-language systems on arXiv.
- Developed a video-to-structured-artefact pipeline using Qwen2.5-VL to extract person detections and attribute hypotheses with schema-based validation.
- Benchmarked interactive inference paths and integrated quantitative evaluation frameworks to ensure semantic output correctness.

PROJECTS

ECG-Analysis — PyTorch, NumPy, Pandas, WFDB, scikit-learn, Cursor

Multi-label 12-lead ECG diagnosis classifier (PTB-XL + Chapman-Shaoxing)

December 2025

- Architected a custom 1D CNN using multi-scale temporal convolutions and Squeeze-and-Excitation attention to classify 500 Hz 12-lead ECGs into 4 diagnostic superclasses (MI, STTC, CD, HYP) via multi-label learning (BCEWithLogits).
- Unified heterogeneous datasets (PTB-XL + Chapman-Shaoxing) with tensor caching, achieving a best validation loss of 0.235.

EvoForge — Rust, Bevy ECS, Rayon, XAI, Python, Cursor

High-performance simulation system with scalable backend architecture

November 2025

- Architected a high-concurrency simulation engine in Rust using the Bevy ECS framework, leveraging data-oriented design (DOD) to manage thousands of entities with parallel system execution and cache-efficient memory layouts.
- Implemented thread-safe simulation logic for organism behaviors, including energy decay, reproduction, and movement, utilizing Rayon for multi-threaded performance and ensuring memory safety without a garbage collector.
- Designed a modular plugin-based architecture to decouple core simulation logic from rendering and telemetry, allowing for reproducible system benchmarks and high-speed headless execution.

Transight — LightGBM, Python, Pandas, FastAPI, Mapbox

Full-stack system for predicting and visualizing real-time transit delays

Hack The Valley: Sept 2025

- Developed an end-to-end transit delay forecasting system using LightGBM, engineering spatio-temporal features from live transit feeds to achieve a 50% improvement in prediction precision over baseline schedules.
- Built a high-throughput asynchronous backend with FastAPI, implementing Pydantic schemas for strict data validation and serving low-latency inference results to a Mapbox GL visualization frontend.
- Containerized the microservice architecture using Docker and automated the deployment lifecycle via GitHub Actions, ensuring 99.9% availability for real-time data ingestion and processing during peak load.

SKILLS

Languages/Frameworks: Python, Java, C++, TypeScript, Rust, React, FastAPI, Node.js

Data & ML: PyTorch, Pandas, NumPy, Data Validation, Model Evaluation

Tools: Docker, AWS, CI/CD (GitHub Actions, CircleCI), SQL, Git